

8 (a) (i) 1.

$$\frac{GM_1M_2}{(R_1 + R_2)^2}$$

2. $M_1R_1\omega^2$

(ii) $\omega = \frac{2\pi}{T} = \frac{2\pi}{1.26 \times 10^8} = 5.0 \times 10^{-8}$

(iii) Force on each of the stars is the same, $\frac{GM_1M_2}{(R_1 + R_2)^2}$

$$\frac{GM_1M_2}{(R_1 + R_2)^2} = M_1R_1\omega^2 = M_2R_2\omega^2$$

Since angular velocity is the same,

$$M_1R_1 = M_2R_2$$

$$\text{Hence } \frac{M_1}{M_2} = \frac{R_2}{R_1}$$

(iv) $R_1 = 0.80 \times 10^{11} \text{ m}, R_2 = 2.4 \times 10^{11} \text{ m}$

(v)

$$\frac{GM_1M_2}{(R_1 + R_2)^2} = M_1R_1\omega^2 \quad [\text{C1}]$$

$$M_2 = \frac{R_1\omega^2(R_1 + R_2)^2}{G}$$

$$= \frac{(0.80 \times 10^{11})(5.0 \times 10^{-8})^2(3.2 \times 10^{11})^2}{6.67 \times 10^{-11}} \quad [\text{M1}]$$

$$= 3.1 \times 10^{29} \quad [\text{A1}]$$

(b) (i) Force on electron $= qE = (1.6 \times 10^{-19})(4000) = 6.4 \times 10^{-16} \text{ N}$

Acceleration of electron $a_y = F / m = (6.4 \times 10^{-16}) / (9.11 \times 10^{-31}) \quad [\text{M1}]$
 $= 7.025 \times 10^{14} \text{ m s}^{-2} \quad [\text{A0}]$

(ii) Time taken = length / horizontal speed $= (5.1 \times 10^{-2}) / 1.7 \times 10^7 = 3.0 \times 10^{-9} \text{ s} \quad [\text{C1}]$

Final vertical component of velocity $v_y = u_y + a_y t = 0 + (7.025 \times 10^{14})(3.0 \times 10^{-9}) = 2.1 \times 10^6 \text{ m s}^{-1} \quad [\text{C1}]$

Final velocity $= \sqrt{v_x^2 + v_y^2} = \sqrt{(1.7 \times 10^7)^2 + (2.1 \times 10^6)^2} = 1.713 \times 10^7 \text{ m s}^{-1}$
 $[\text{A1}]$

$\theta = \tan^{-1} \left(\frac{2.1 \times 10^6}{1.7 \times 10^7} \right) = 7.04^\circ \quad [\text{A1}]$

(iii) Deflection in opposite direction [A0] due to different sign of charge [M1]

Path is less curved [A0] because proton is less strongly deflected. Proton has same magnitude of charge as electron, experiences same magnitude of electric force, but proton has much larger mass $\rightarrow$ acceleration is much smaller [M1]

(c) (i) Path in B-field is circular, Path in E-field is parabolic. [B1]

Circular – because magnetic force is (constant in magnitude and) always perpendicular to the velocity, so provides constant centripetal force toward a centre, causing uniform circular motion [B1]

Parabolic – because electric force is constant (in magnitude and direction) and in only one direction (perpendicular to the initial velocity), so acceleration in one direction and constant velocity in perpendicular direction. [B1]

(ii) Because centripetal force only changes direction and not magnitude of velocity [M1]

(Electron is in uniform circular motion),

Final speed of electron is the same as initial speed $= 5.1 \times 10^7 \text{ m s}^{-1}$

Final speed of electron in (b) has a higher final speed than electron in (c) [A1]

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9 (a) Not affected by external factors / conditions [B1]